Causal Analysis Report

bpic12-a.xes.gz-im

Decision points:	54594
Choice sets:	7
Transitions fitted:	11
Significant (p<0.05):	2
Generated:	2026-06-22 08:47

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

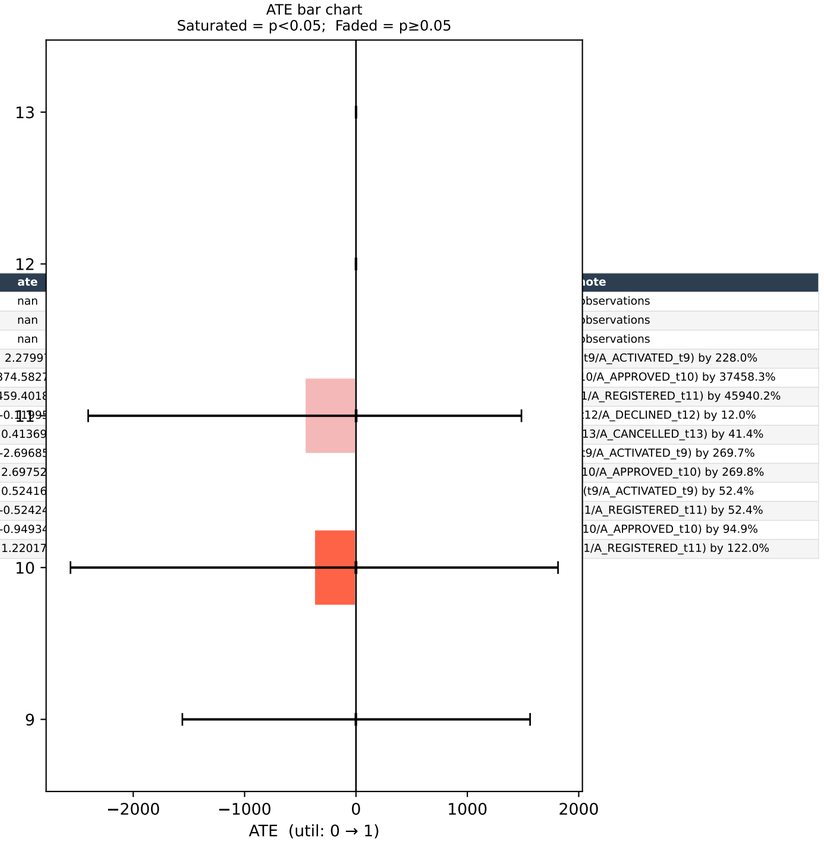
(no data)

Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

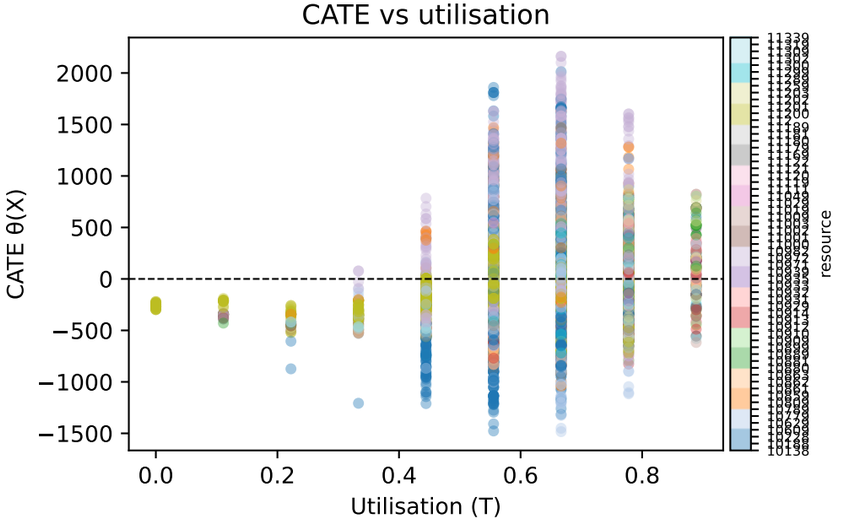
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(2, 3)	('τ_t2', 'A_PREACCEPTED_t3')	7367	3	—	—	—	nan	nan	—	nan	—	too few observations
(4, 5)	('τ_t4', 'A_ACCEPTED_t5')	5113	5	—	—	—	nan	nan	—	nan	—	too few observations
(6, 7)	('τ_t6', 'A_FINALIZED_t7')	5015	7	—	—	—	nan	nan	—	nan	—	too few observations
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	9	A_ACTIVATED_t9	A_APPROVED_t10, A_CANCELLED_t13, A_DECLINED_t12, A_REGISTERED_t11	-0.222601	2.346304	[-4.8214, 4.3762]	0.9244	False	θ=-0.2226: 1-unit ↑ util ↓ P(t9/A_ACTIVATED_t9) by 22.3%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	10	A_APPROVED_t10	A_ACTIVATED_t9, A_CANCELLED_t13, A_DECLINED_t12, A_REGISTERED_t11	0.52222	3.021994	[-5.4009, 6.4453]	0.8628	False	θ=+0.5222: 1-unit ↑ util ↑ P(t10/A_APPROVED_t10) by 52.2%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	11	A_REGISTERED_t11	A_ACTIVATED_t9, A_APPROVED_t10, A_CANCELLED_t13, A_DECLINED_t12	0.237885	2.682848	[-5.0205, 5.4963]	0.9293	False	θ=+0.2379: 1-unit ↑ util ↑ P(t11/A_REGISTERED_t11) by 23.8%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	12	A_DECLINED_t12	A_ACTIVATED_t9, A_APPROVED_t10, A_CANCELLED_t13, A_REGISTERED_t11	0.215279	0.28318	[-0.3398, 0.7703]	0.4471	False	θ=+0.2153: 1-unit ↑ util ↑ P(t12/A_DECLINED_t12) by 21.5%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	13	A_CANCELLED_t13	A_ACTIVATED_t9, A_APPROVED_t10, A_DECLINED_t12, A_REGISTERED_t11	0.494409	0.24941	[0.0056, 0.9833]	0.0474	True	θ=+0.4944: 1-unit ↑ util ↑ P(t13/A_CANCELLED_t13) by 49.4%
(9, 10)	('A_ACTIVATED_t9', 'A_APPROVED_t10')	715	9, 10	9	A_ACTIVATED_t9	A_APPROVED_t10	-0.319779	0.95278	[-2.1872, 1.5477]	0.7372	False	θ=-0.3198: 1-unit ↑ util ↓ P(t9/A_ACTIVATED_t9) by 32.0%
(9, 10)	('A_ACTIVATED_t9', 'A_APPROVED_t10')	715	9, 10	10	A_APPROVED_t10	A_ACTIVATED_t9	0.319779	0.95278	[-1.5477, 2.1872]	0.7372	False	θ=+0.3198: 1-unit ↑ util ↑ P(t10/A_APPROVED_t10) by 32.0%
(9, 11)	('A_ACTIVATED_t9', 'A_REGISTERED_t11')	1055	9, 11	9	A_ACTIVATED_t9	A_REGISTERED_t11	0.302604	1.236002	[-2.1200, 2.7252]	0.8066	False	θ=+0.3026: 1-unit ↑ util ↑ P(t9/A_ACTIVATED_t9) by 30.3%
(9, 11)	('A_ACTIVATED_t9', 'A_REGISTERED_t11')	1055	9, 11	11	A_REGISTERED_t11	A_ACTIVATED_t9	-0.302604	1.236002	[-2.7252, 2.1200]	0.8066	False	θ=-0.3026: 1-unit ↑ util ↓ P(t11/A_REGISTERED_t11) by 30.3%
(10, 11)	('A_APPROVED_t10', 'A_REGISTERED_t11')	476	10, 11	10	A_APPROVED_t10	A_REGISTERED_t11	0.224918	0.923228	[-1.5846, 2.0344]	0.8075	False	θ=+0.2249: 1-unit ↑ util ↑ P(t10/A_APPROVED_t10) by 22.5%
(10, 11)	('A_APPROVED_t10', 'A_REGISTERED_t11')	476	10, 11	11	A_REGISTERED_t11	A_APPROVED_t10	-0.224918	0.923228	[-2.0344, 1.5846]	0.8075	False	θ=-0.2249: 1-unit ↑ util ↓ P(t11/A_REGISTERED_t11) by 22.5%

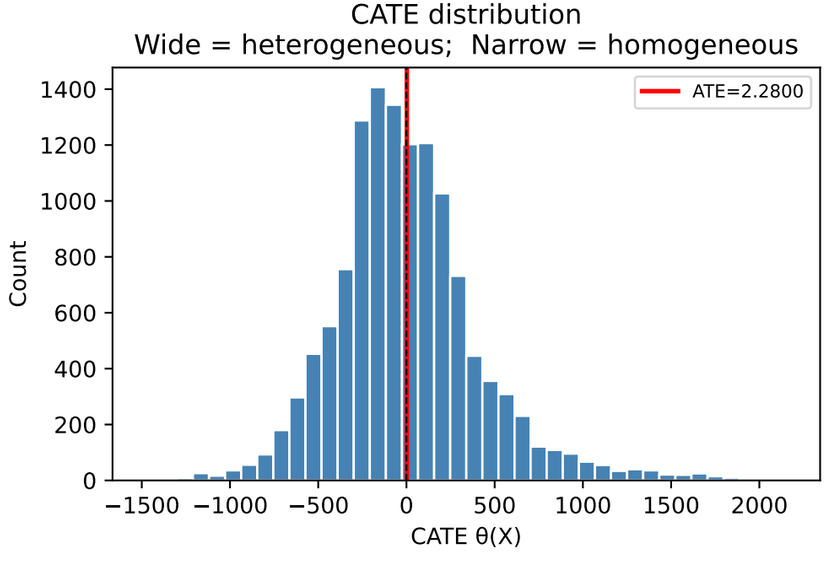
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

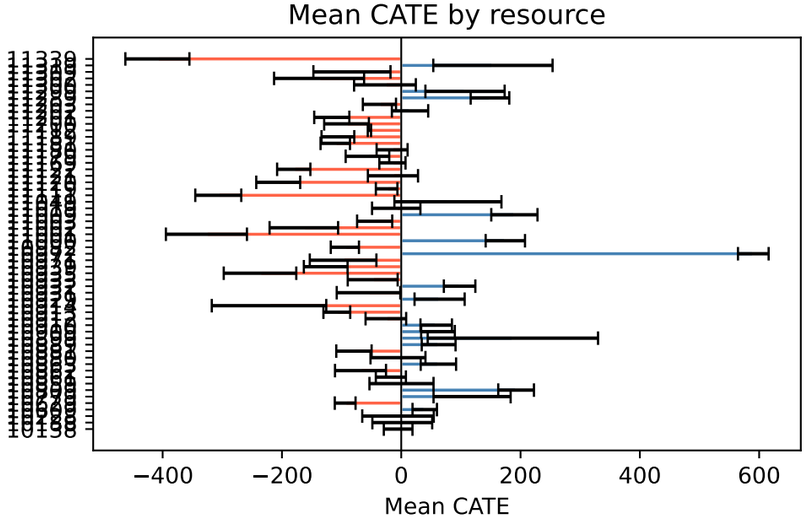
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ote
(2, 3)	('τ_t2', 'A_PREACCEPTED_t3')	7367	3	—	—	—	nan	bservations
(4, 5)	('τ_t4', 'A_ACCEPTED_t5')	5113	5	—	—	—	nan	bservations
(6, 7)	('τ_t6', 'A_FINALIZED_t7')	5015	7	—	—	—	nan	bservations
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	9	A_ACTIVATED_t9	A_APPROVED_t10, A_CANCELLED_t13, A_DECLINED_t12, A_REGISTERED_t11	2.2799	t9/A_ACTIVATED_t9) by 228.0%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	10	A_APPROVED_t10	A_ACTIVATED_t9, A_CANCELLED_t13, A_DECLINED_t12, A_REGISTERED_t11	-374.5827	0/A_APPROVED_t10) by 37458.3%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	11	A_REGISTERED_t11	A_ACTIVATED_t9, A_APPROVED_t10, A_CANCELLED_t13, A_DECLINED_t12	-459.4018	1/A_REGISTERED_t11) by 45940.2%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	12	A_DECLINED_t12	A_ACTIVATED_t9, A_APPROVED_t10, A_CANCELLED_t13, A_REGISTERED_t11	-0.1199	12/A_DECLINED_t12) by 12.0%
(8, 9, 10, 11, 12, 13)	('τ_t8', 'A_ACTIVATED_t9', 'A_APPROVED_t10', 'A_REGISTERED_t11', 'A_DECLINED_t12', 'A_CANCELLED_t13')	12688	9, 10, 11, 12, 13	13	A_CANCELLED_t13	A_ACTIVATED_t9, A_APPROVED_t10, A_DECLINED_t12, A_REGISTERED_t11	0.41369	13/A_CANCELLED_t13) by 41.4%
(9, 10)	('A_ACTIVATED_t9', 'A_APPROVED_t10')	715	9, 10	9	A_ACTIVATED_t9	A_APPROVED_t10	-2.6968	t9/A_ACTIVATED_t9) by 269.7%
(9, 10)	('A_ACTIVATED_t9', 'A_APPROVED_t10')	715	9, 10	10	A_APPROVED_t10	A_ACTIVATED_t9	2.69752	10/A_APPROVED_t10) by 269.8%
(9, 11)	('A_ACTIVATED_t9', 'A_REGISTERED_t11')	1055	9, 11	9	A_ACTIVATED_t9	A_REGISTERED_t11	0.52416	(t9/A_ACTIVATED_t9) by 52.4%
(9, 11)	('A_ACTIVATED_t9', 'A_REGISTERED_t11')	1055	9, 11	11	A_REGISTERED_t11	A_ACTIVATED_t9	-0.5242	1/A_REGISTERED_t11) by 52.4%
(10, 11)	('A_APPROVED_t10', 'A_REGISTERED_t11')	476	10, 11	10	A_APPROVED_t10	A_REGISTERED_t11	-0.9493	10/A_APPROVED_t10) by 94.9%
(10, 11)	('A_APPROVED_t10', 'A_REGISTERED_t11')	476	10, 11	11	A_REGISTERED_t11	A_APPROVED_t10	1.22017	1/A_REGISTERED_t11) by 122.0%

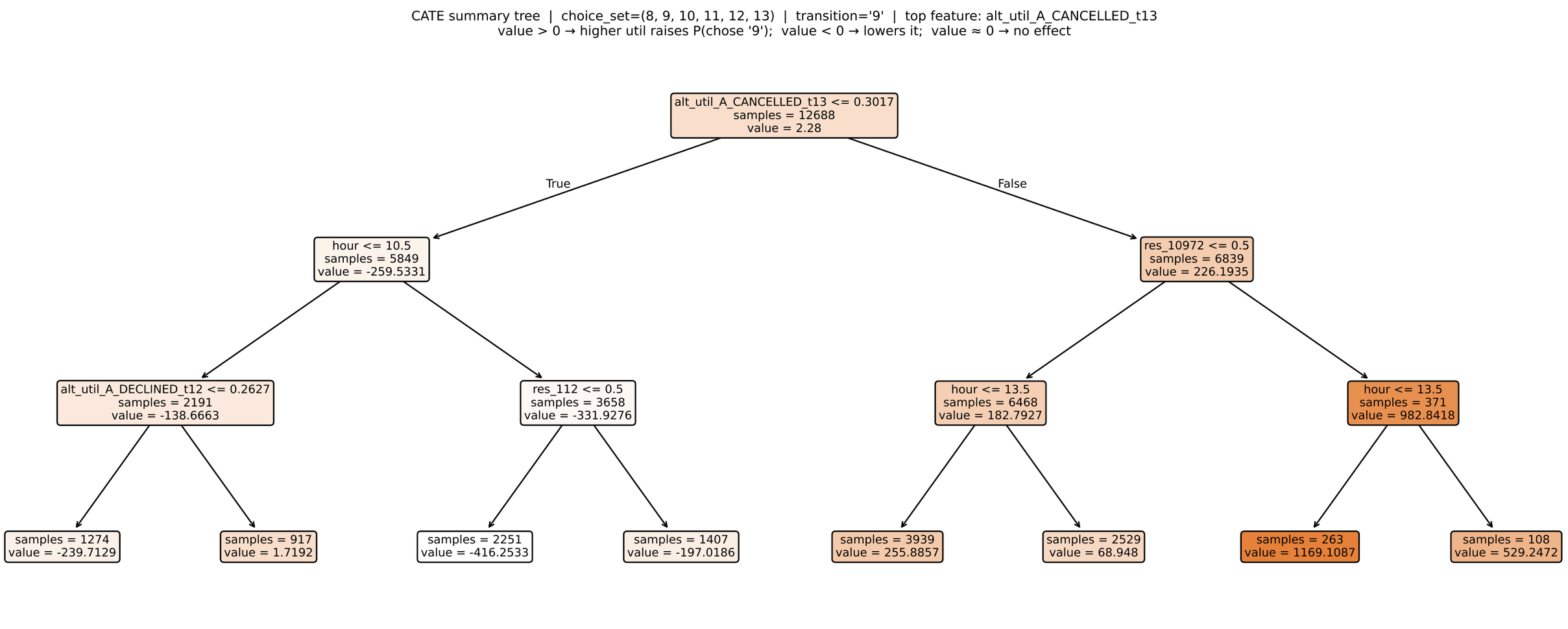


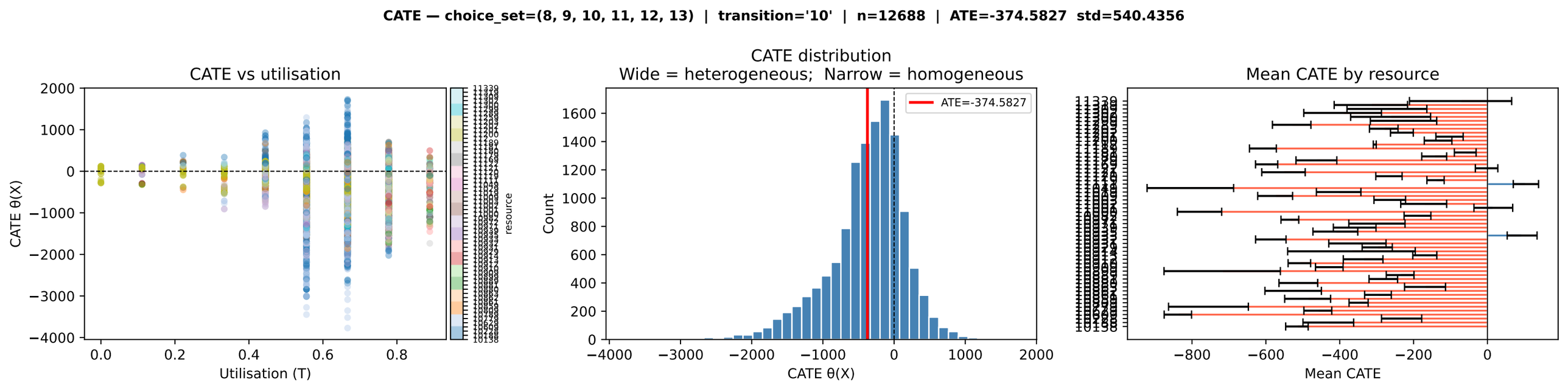
CATE — choice_set=(8, 9, 10, 11, 12, 13) | transition='9' | n=12688 | ATE=2.2800 std=412.3961

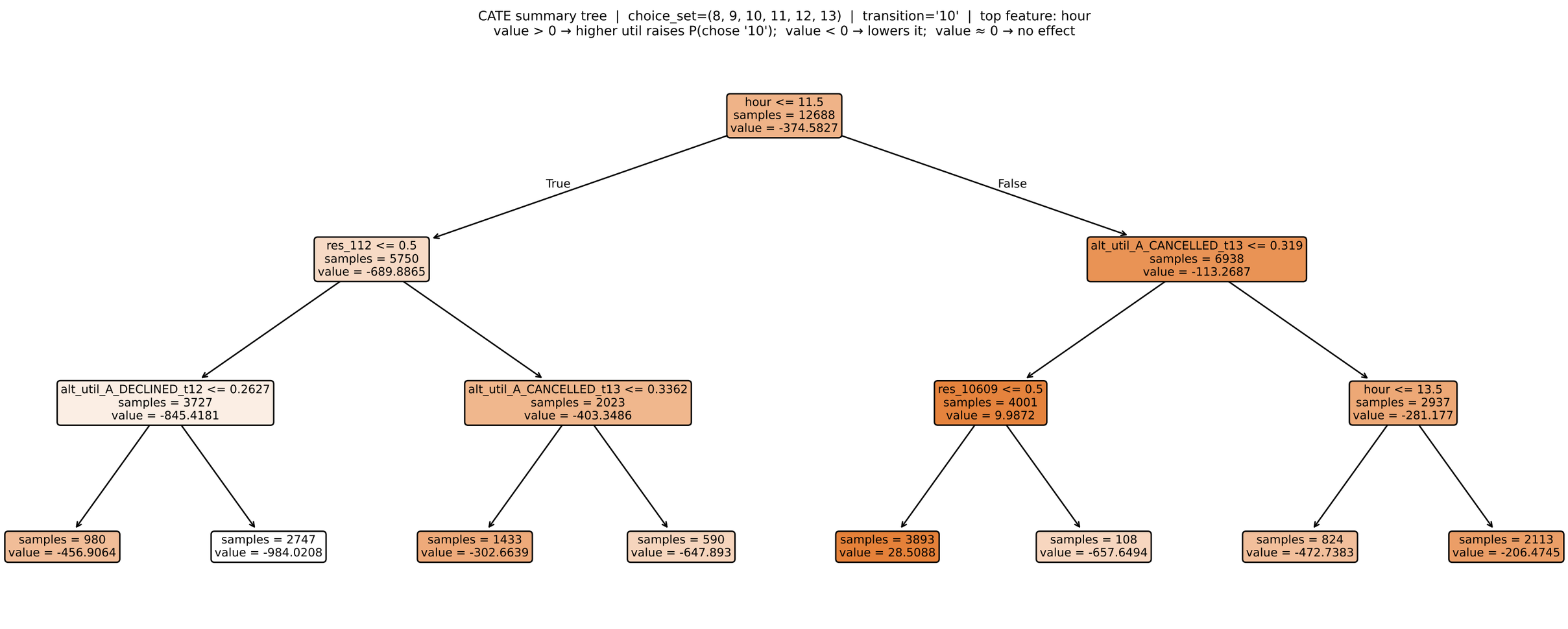


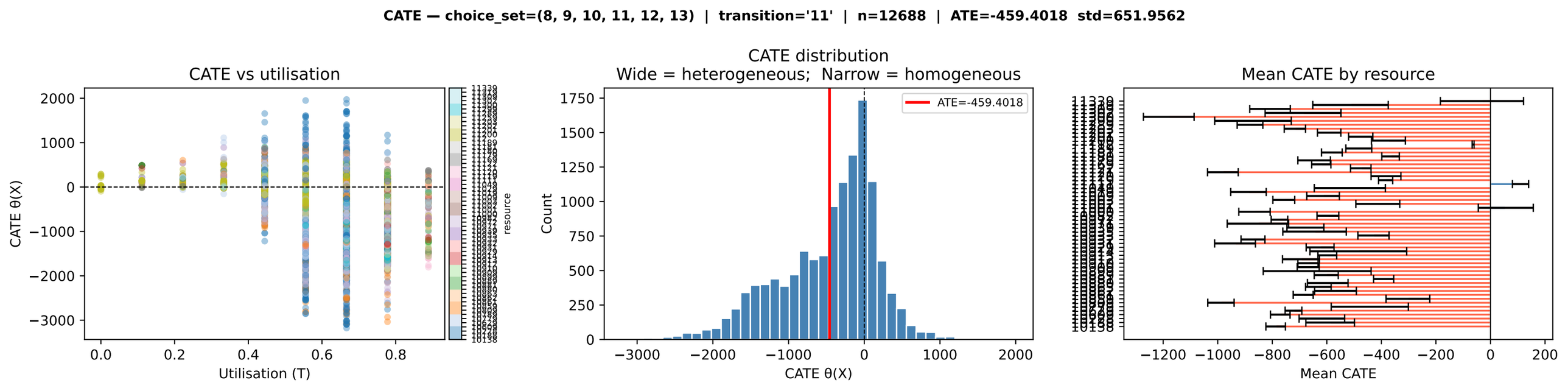


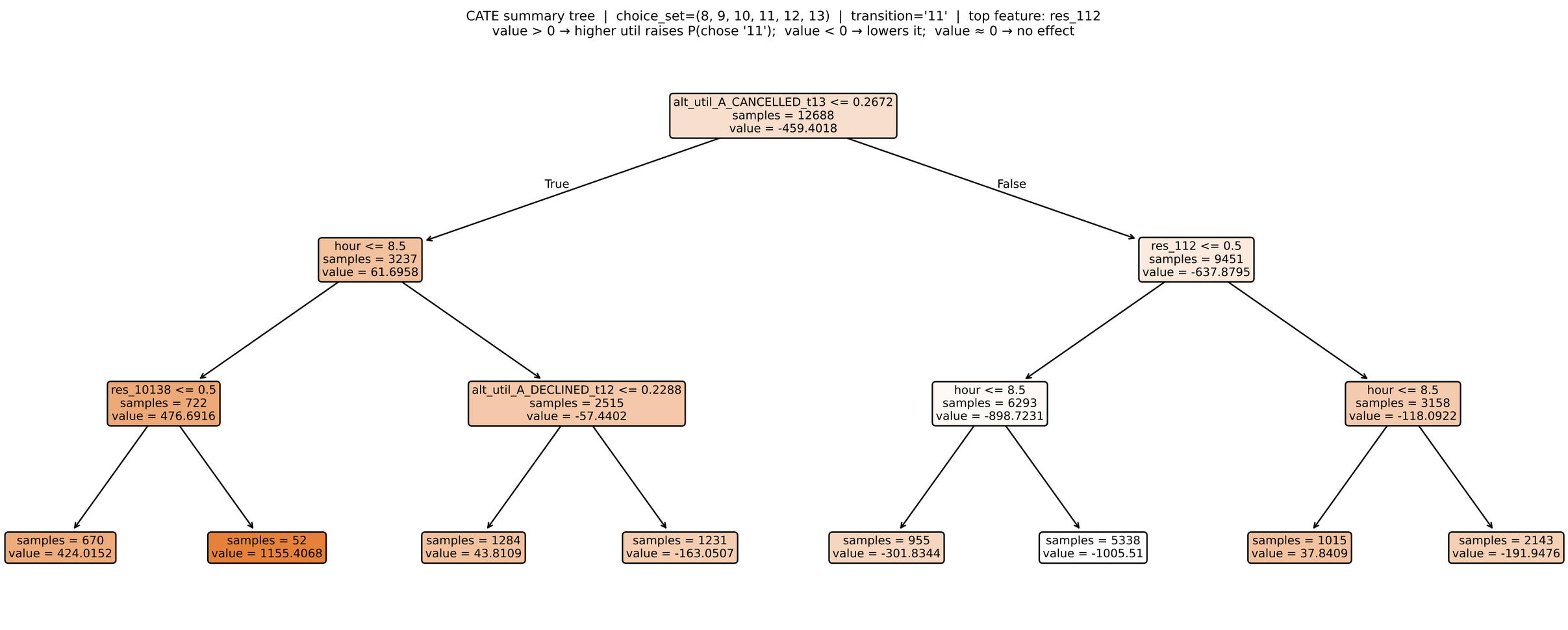



CATE summary tree | choice_set=(8, 9, 10, 11, 12, 13) | transition='9' | top feature: alt_util_A_CANCELLED_t13
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_CANCELLED_t13 <= 0.3017
samples = 12688
value = 2.28
True
False
hour <= 10.5
samples = 5849
value = -259.5331
res_10972 <= 0.5
samples = 6839
value = 226.1935
alt_util_A_DECLINED_t12 <= 0.2627
samples = 2191
value = -138.6663
res_112 <= 0.5
samples = 3658
value = -331.9276
hour <= 13.5
samples = 6468
value = 182.7927
hour <= 13.5
samples = 371
value = 982.8418
samples = 1274
value = -239.7129
samples = 917
value = 1.7192
samples = 2251
value = -416.2533
samples = 1407
value = -197.0186
samples = 3939
value = 255.8857
samples = 2529
value = 68.948
samples = 263
value = 1169.1087
samples = 108
value = 529.2472

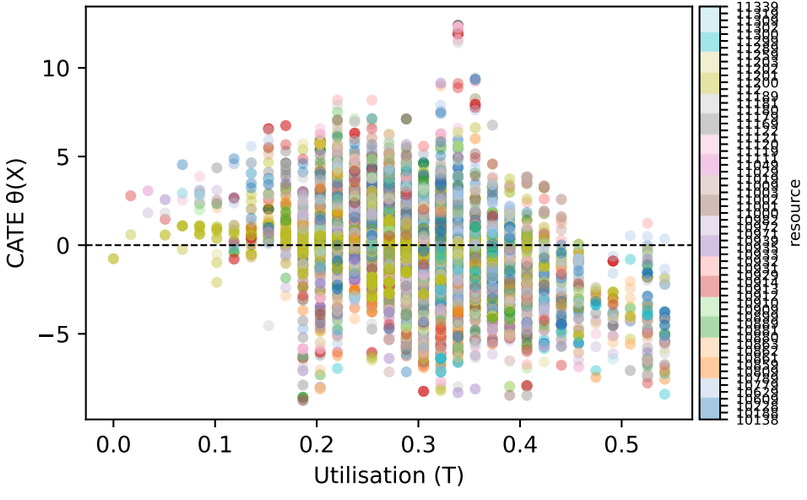

CATE — choice_set=(8, 9, 10, 11, 12, 13) | transition='10' | n=12688 | ATE=-374.5827 std=540.4356
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-374.5827
Mean CATE by resource
Mean CATE

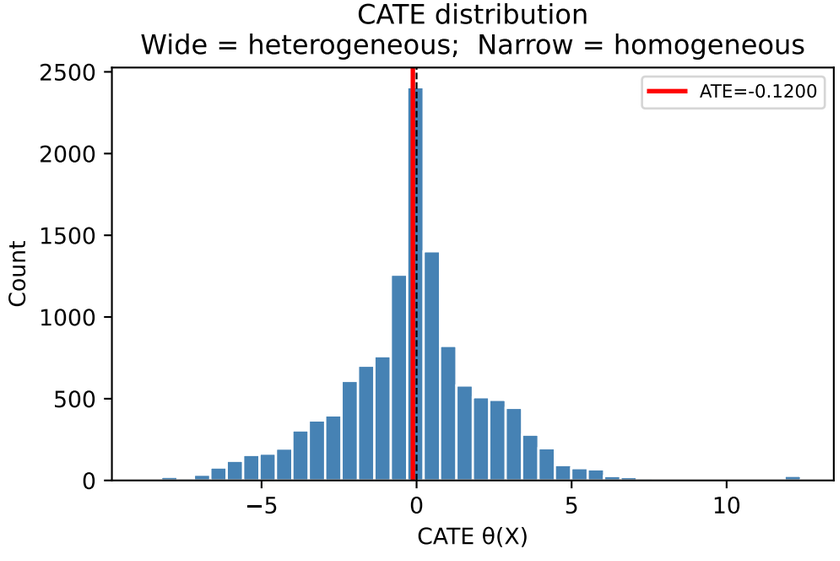

CATE summary tree | choice_set=(8, 9, 10, 11, 12, 13) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 11.5
samples = 12688
value = -374.5827
True
False
res_112 <= 0.5
samples = 5750
value = -689.8865
alt_util_A_CANCELLED_t13 <= 0.319
samples = 6938
value = -113.2687
alt_util_A_DECLINED_t12 <= 0.2627
samples = 3727
value = -845.4181
alt_util_A_CANCELLED_t13 <= 0.3362
samples = 2023
value = -403.3486
res_10609 <= 0.5
samples = 4001
value = 9.9872
hour <= 13.5
samples = 2937
value = -281.177
samples = 980
value = -456.9064
samples = 2747
value = -984.0208
samples = 1433
value = -302.6639
samples = 590
value = -647.893
samples = 3893
value = 28.5088
samples = 108
value = -657.6494
samples = 824
value = -472.7383
samples = 2113
value = -206.4745

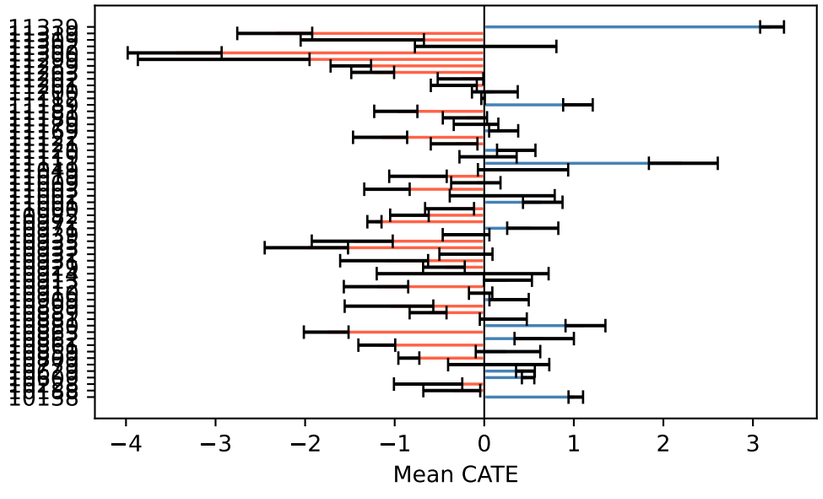

CATE — choice_set=(8, 9, 10, 11, 12, 13) | transition='11' | n=12688 | ATE=-459.4018 std=651.9562
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-459.4018
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

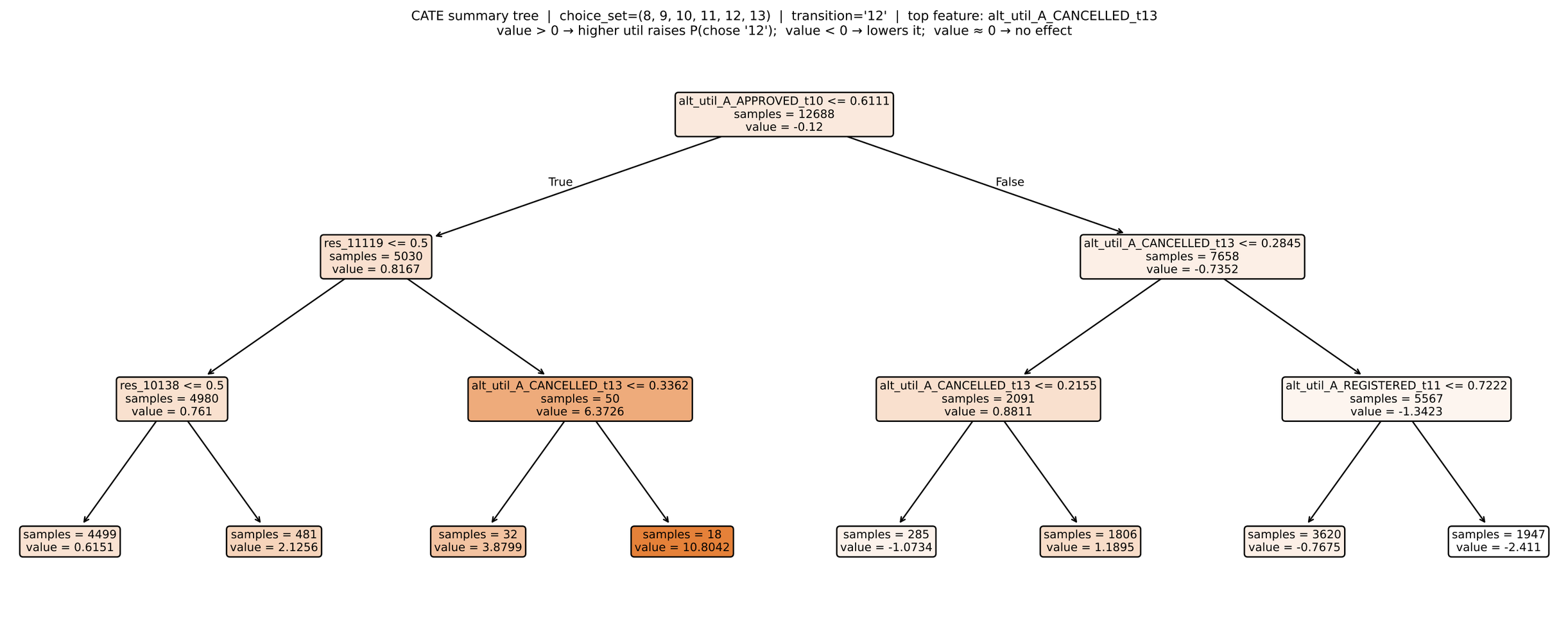

CATE summary tree | choice_set=(8, 9, 10, 11, 12, 13) | transition='11' | top feature: res_112
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_CANCELLED_t13 <= 0.2672
samples = 12688
value = -459.4018
True
False
hour <= 8.5
samples = 3237
value = 61.6958
res_112 <= 0.5
samples = 9451
value = -637.8795
res_10138 <= 0.5
samples = 722
value = 476.6916
alt_util_A_DECLINED_t12 <= 0.2288
samples = 2515
value = -57.4402
hour <= 8.5
samples = 6293
value = -898.7231
hour <= 8.5
samples = 3158
value = -118.0922
samples = 670
value = 424.0152
samples = 52
value = 1155.4068
samples = 1284
value = 43.8109
samples = 1231
value = -163.0507
samples = 955
value = -301.8344
samples = 5338
value = -1005.51
samples = 1015
value = 37.8409
samples = 2143
value = -191.9476

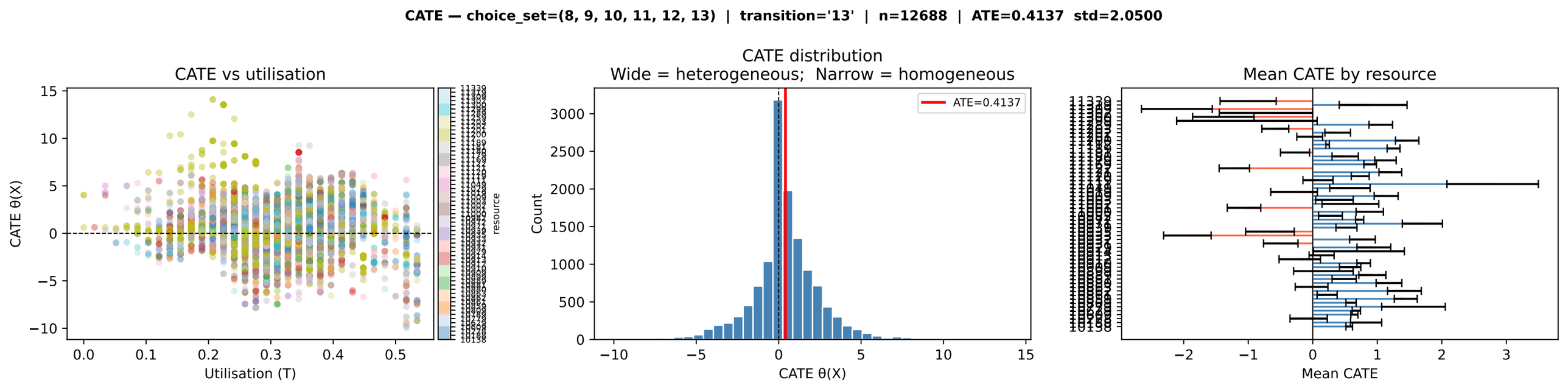
CATE — choice_set=(8, 9, 10, 11, 12, 13) | transition='12' | n=12688 | ATE=-0.1200 std=2.4449

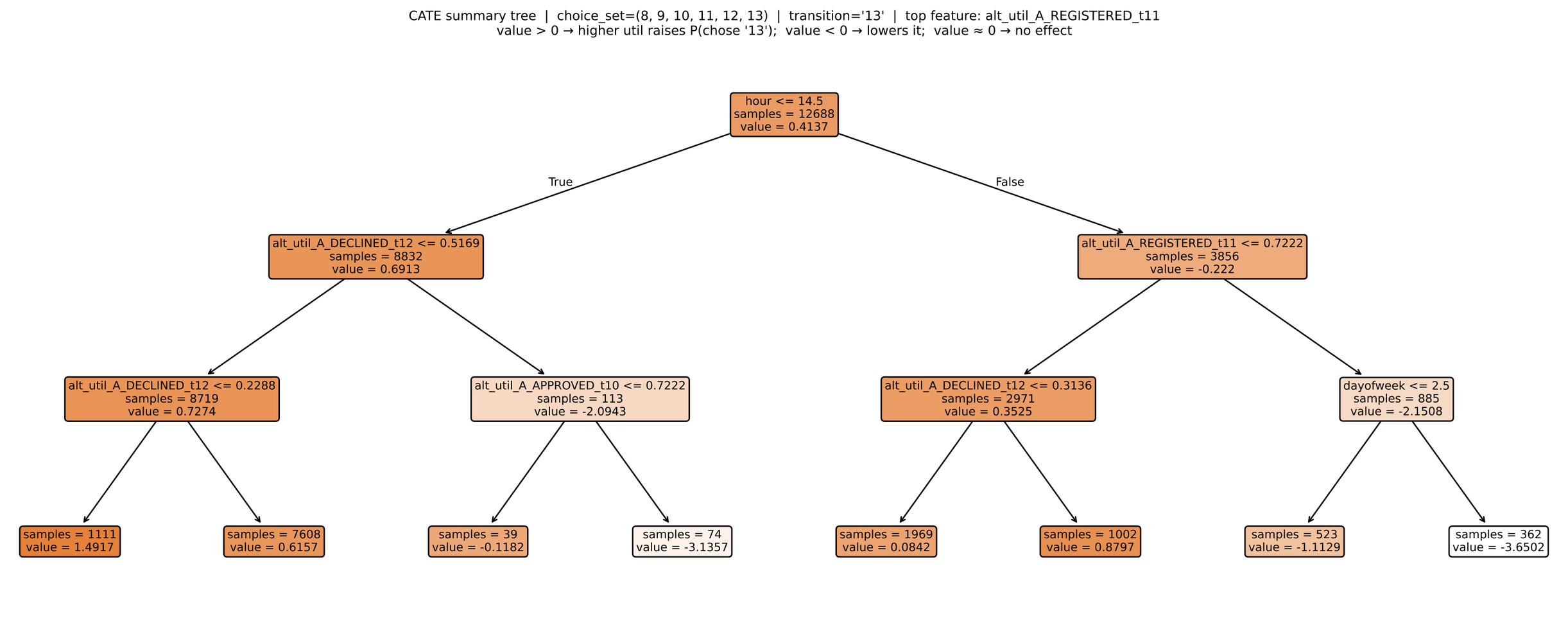


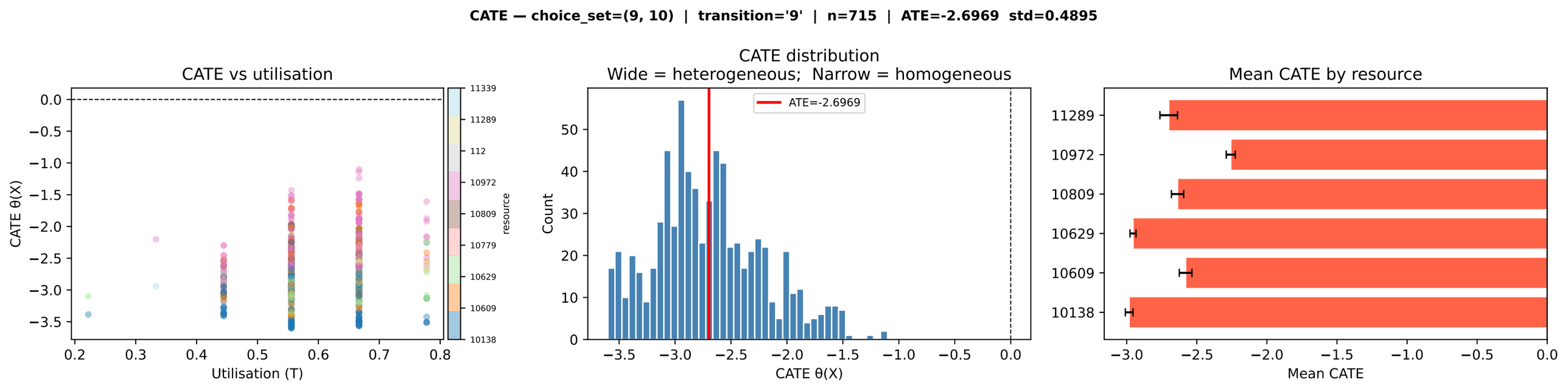


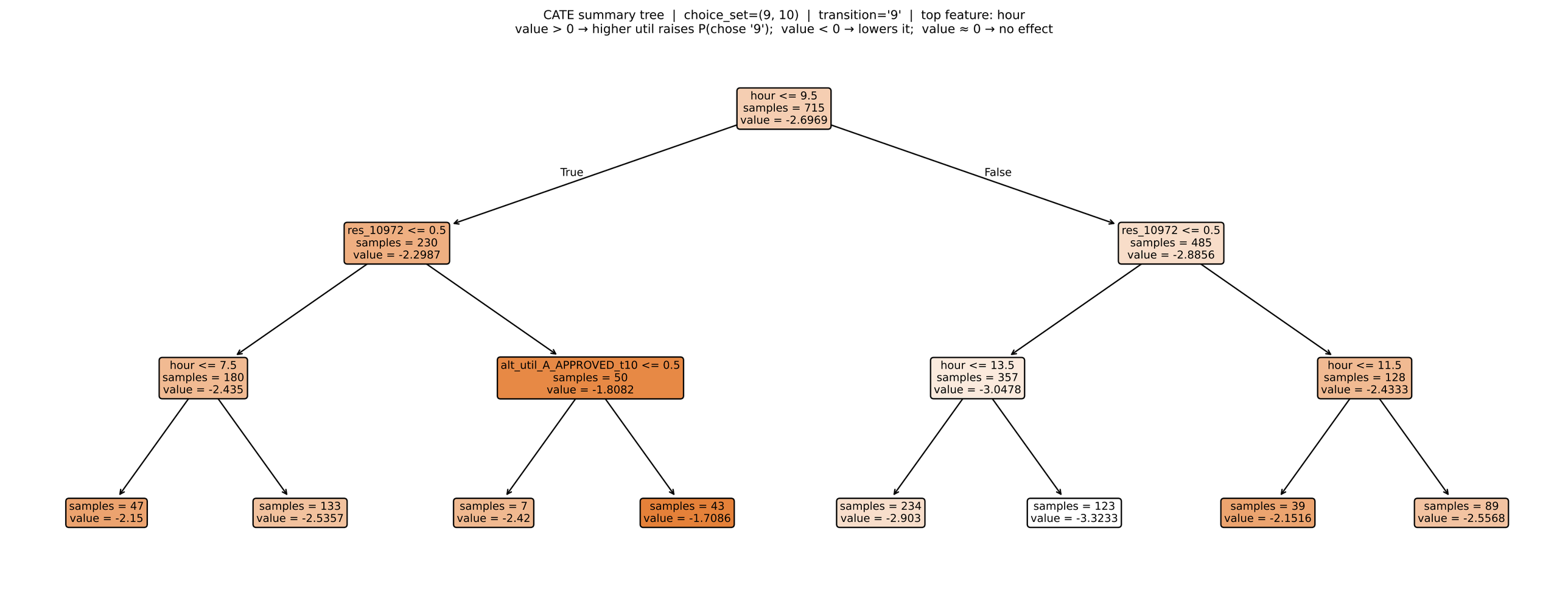


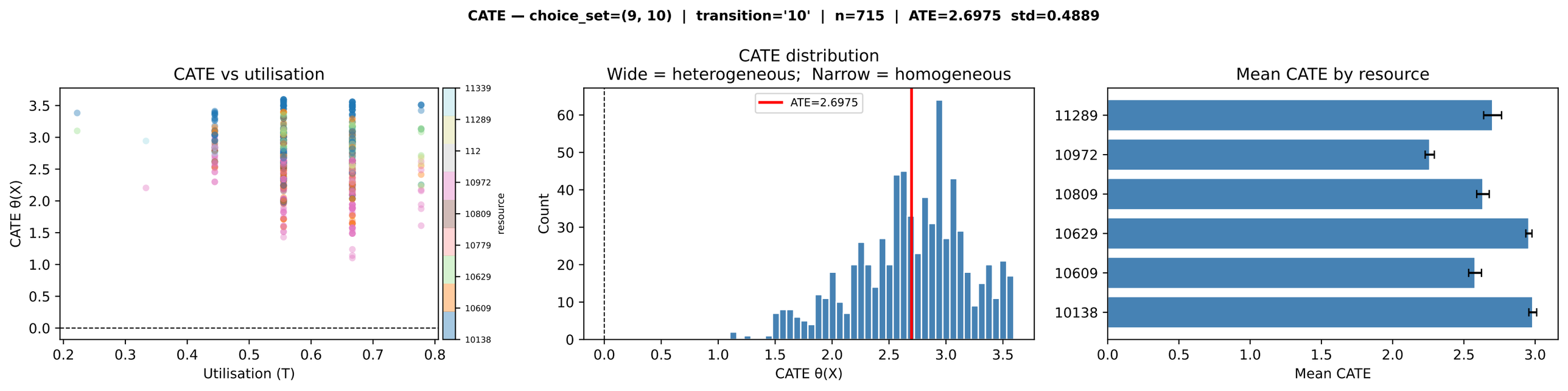

CATE summary tree | choice_set=(8, 9, 10, 11, 12, 13) | transition='12' | top feature: alt_util_A_CANCELLED_t13
value > 0 → higher util raises P(chose '12'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_APPROVED_t10 <= 0.6111
samples = 12688
value = -0.12
True
False
res_11119 <= 0.5
samples = 5030
value = 0.8167
alt_util_A_CANCELLED_t13 <= 0.2845
samples = 7658
value = -0.7352
res_10138 <= 0.5
samples = 4980
value = 0.761
alt_util_A_CANCELLED_t13 <= 0.3362
samples = 50
value = 6.3726
alt_util_A_CANCELLED_t13 <= 0.2155
samples = 2091
value = 0.8811
alt_util_A_REGISTERED_t11 <= 0.7222
samples = 5567
value = -1.3423
samples = 4499
value = 0.6151
samples = 481
value = 2.1256
samples = 32
value = 3.8799
samples = 18
value = 10.8042
samples = 285
value = -1.0734
samples = 1806
value = 1.1895
samples = 3620
value = -0.7675
samples = 1947
value = -2.411

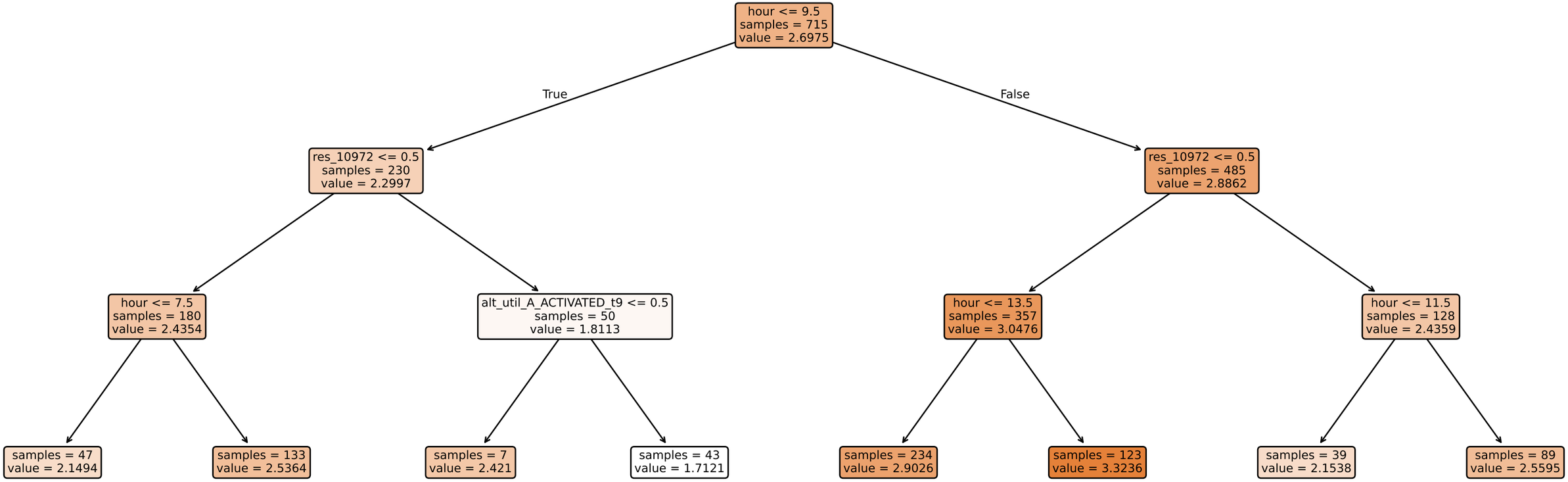

CATE — choice_set=(8, 9, 10, 11, 12, 13) | transition='13' | n=12688 | ATE=0.4137 std=2.0500
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.4137
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(8, 9, 10, 11, 12, 13) | transition='13' | top feature: alt_util_A_REGISTERED_t11
value > 0 → higher util raises P(chose '13'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 14.5
samples = 12688
value = 0.4137
True
False
alt_util_A_DECLINED_t12 <= 0.5169
samples = 8832
value = 0.6913
alt_util_A_REGISTERED_t11 <= 0.7222
samples = 3856
value = -0.222
alt_util_A_DECLINED_t12 <= 0.2288
samples = 8719
value = 0.7274
alt_util_A_APPROVED_t10 <= 0.7222
samples = 113
value = -2.0943
alt_util_A_DECLINED_t12 <= 0.3136
samples = 2971
value = 0.3525
dayofweek <= 2.5
samples = 885
value = -2.1508
samples = 1111
value = 1.4917
samples = 7608
value = 0.6157
samples = 39
value = -0.1182
samples = 74
value = -3.1357
samples = 1969
value = 0.0842
samples = 1002
value = 0.8797
samples = 523
value = -1.1129
samples = 362
value = -3.6502

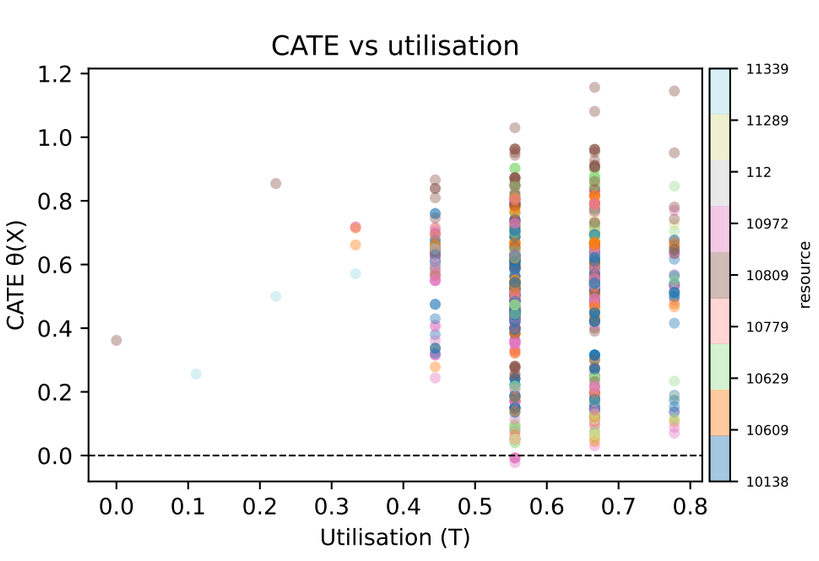

CATE — choice_set=(9, 10) | transition='9' | n=715 | ATE=-2.6969 std=0.4895
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10779
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.6969
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

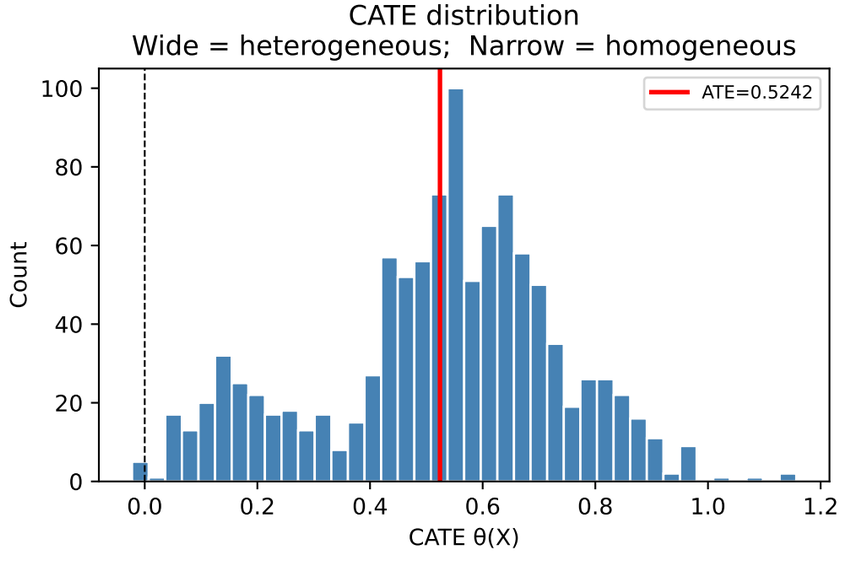

CATE summary tree | choice_set=(9, 10) | transition='9' | top feature: hour
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 9.5
samples = 715
value = -2.6969
True
False
res_10972 <= 0.5
samples = 230
value = -2.2987
res_10972 <= 0.5
samples = 485
value = -2.8856
hour <= 7.5
samples = 180
value = -2.435
alt_util_A_APPROVED_t10 <= 0.5
samples = 50
value = -1.8082
hour <= 13.5
samples = 357
value = -3.0478
hour <= 11.5
samples = 128
value = -2.4333
samples = 47
value = -2.15
samples = 133
value = -2.5357
samples = 7
value = -2.42
samples = 43
value = -1.7086
samples = 234
value = -2.903
samples = 123
value = -3.3233
samples = 39
value = -2.1516
samples = 89
value = -2.5568

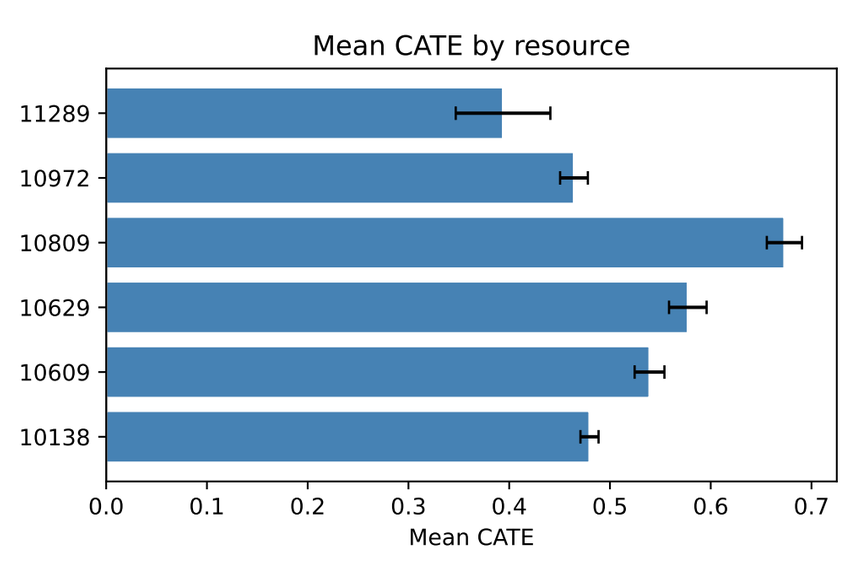

CATE — choice_set=(9, 10) | transition='10' | n=715 | ATE=2.6975 std=0.4889
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10779
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=2.6975
Count
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE

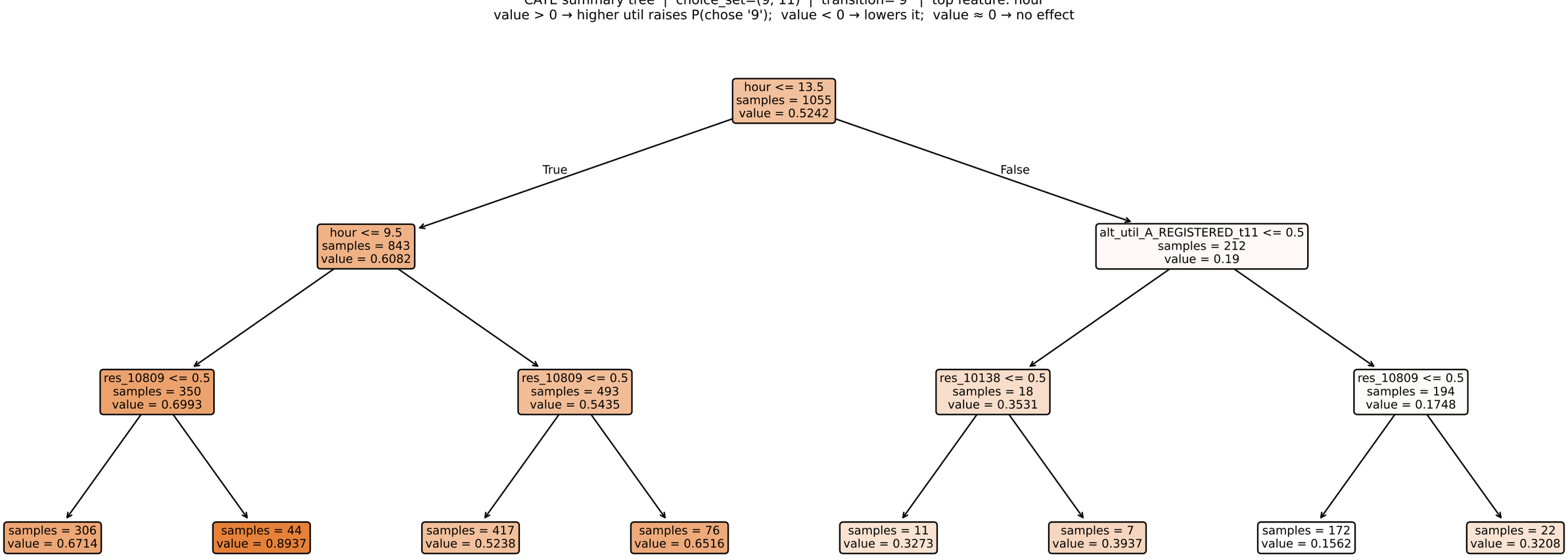

CATE summary tree | choice_set=(9, 10) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 9.5
samples = 715
value = 2.6975
True
False
res_10972 <= 0.5
samples = 230
value = 2.2997
res_10972 <= 0.5
samples = 485
value = 2.8862
hour <= 7.5
samples = 180
value = 2.4354
alt_util_A_ACTIVATED_t9 <= 0.5
samples = 50
value = 1.8113
hour <= 13.5
samples = 357
value = 3.0476
hour <= 11.5
samples = 128
value = 2.4359
samples = 47
value = 2.1494
samples = 133
value = 2.5364
samples = 7
value = 2.421
samples = 43
value = 1.7121
samples = 234
value = 2.9026
samples = 123
value = 3.3236
samples = 39
value = 2.1538
samples = 89
value = 2.5595

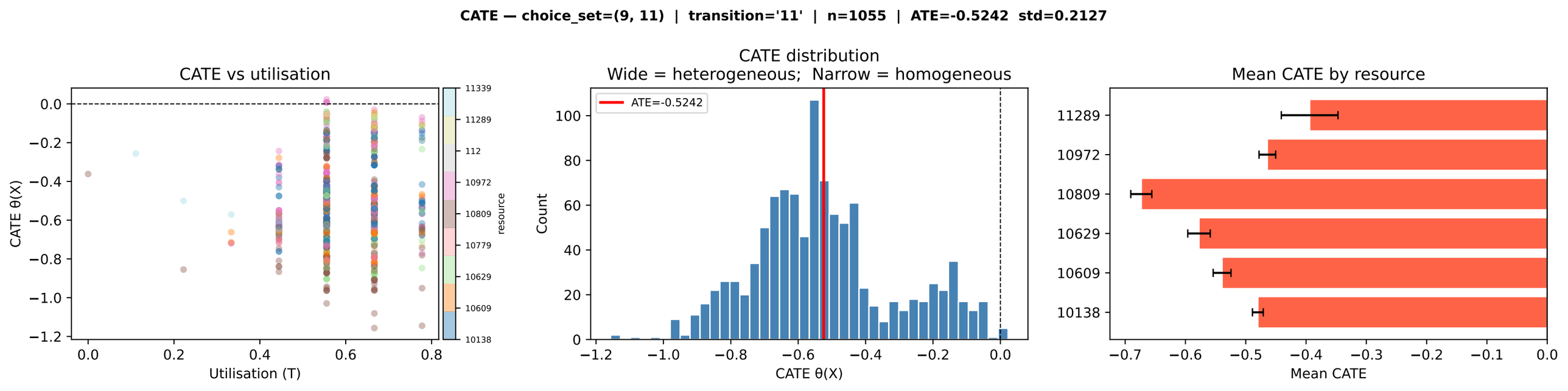
CATE — choice_set=(9, 11) | transition='9' | n=1055 | ATE=0.5242 std=0.2125

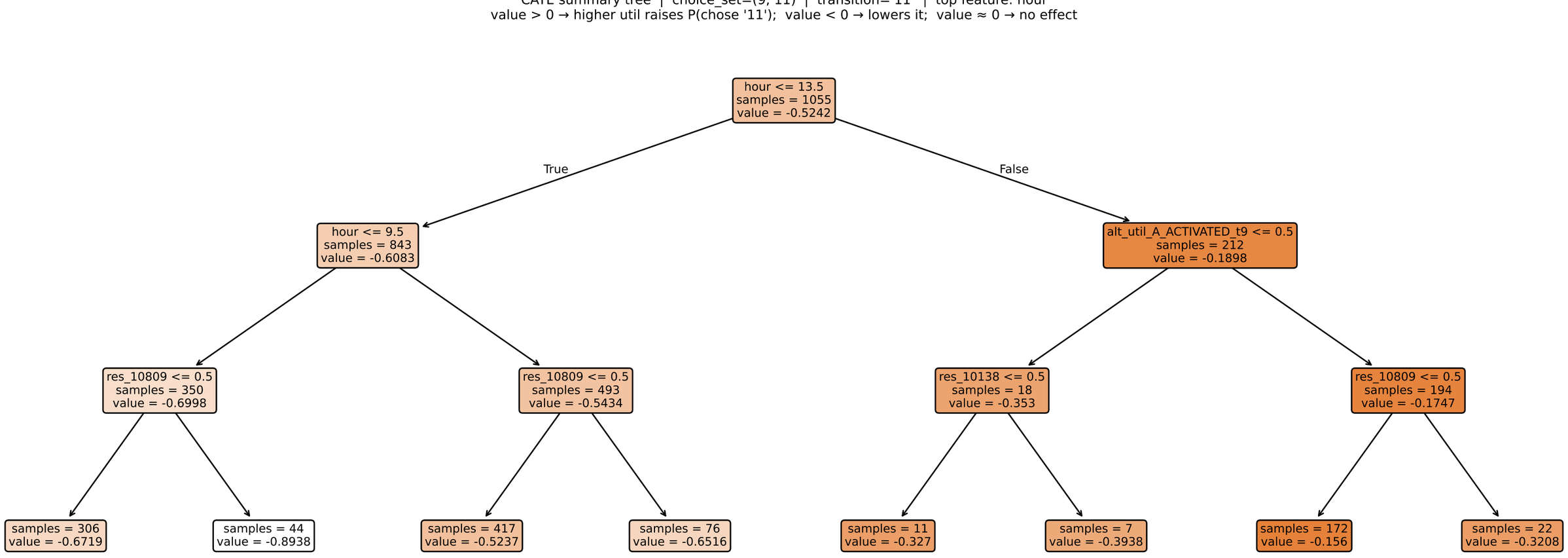




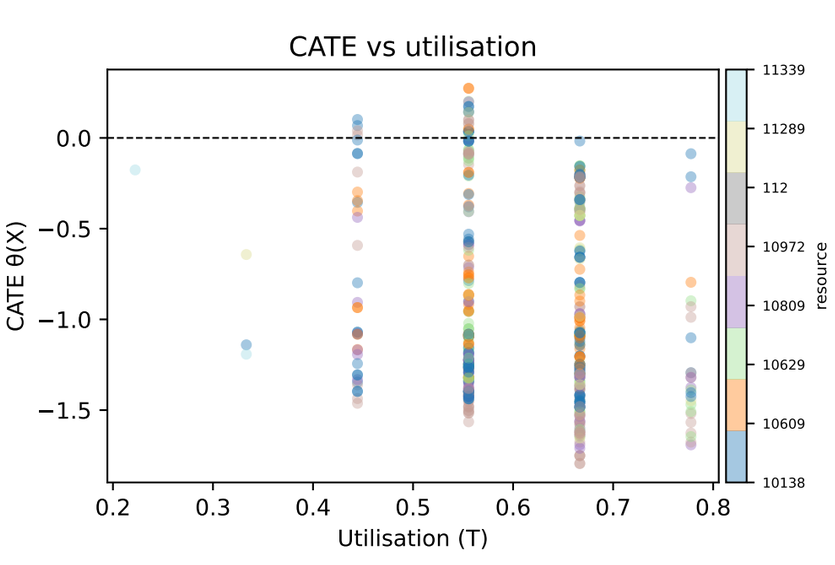


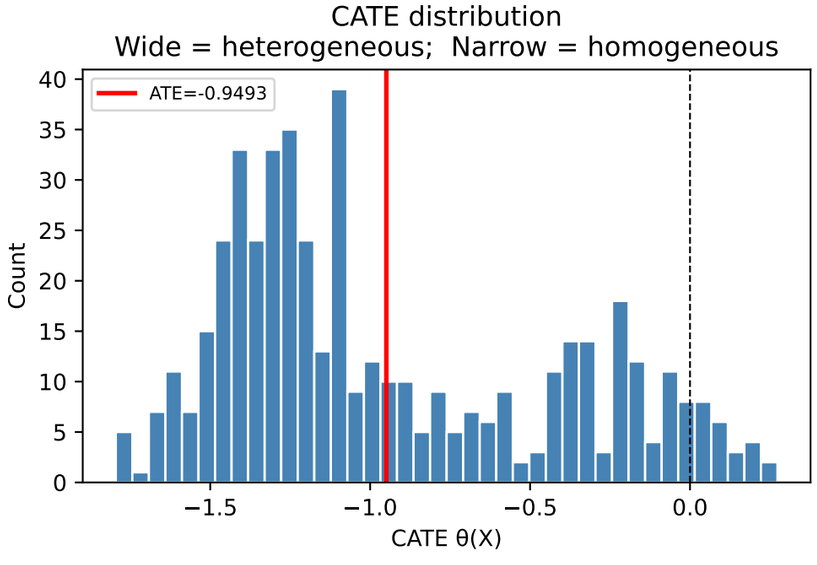

CATE summary tree | choice_set=(9, 11) | transition='9' | top feature: hour
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 13.5
samples = 1055
value = 0.5242
True
False
hour <= 9.5
samples = 843
value = 0.6082
alt_util_A_REGISTERED_t11 <= 0.5
samples = 212
value = 0.19
res_10809 <= 0.5
samples = 350
value = 0.6993
res_10809 <= 0.5
samples = 493
value = 0.5435
res_10138 <= 0.5
samples = 18
value = 0.3531
res_10809 <= 0.5
samples = 194
value = 0.1748
samples = 306
value = 0.6714
samples = 44
value = 0.8937
samples = 417
value = 0.5238
samples = 76
value = 0.6516
samples = 11
value = 0.3273
samples = 7
value = 0.3937
samples = 172
value = 0.1562
samples = 22
value = 0.3208

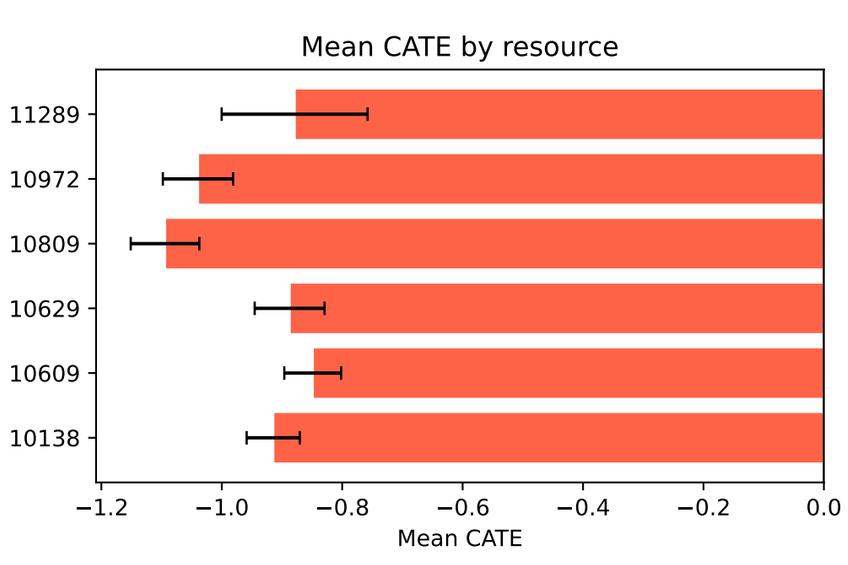

CATE — choice_set=(9, 11) | transition='11' | n=1055 | ATE=-0.5242 std=0.2127
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
11339
11289
112
10972
10809
10779
10629
10609
10138
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.5242
Count
CATE θ(X)
Mean CATE by resource
11289
10972
10809
10629
10609
10138
Mean CATE

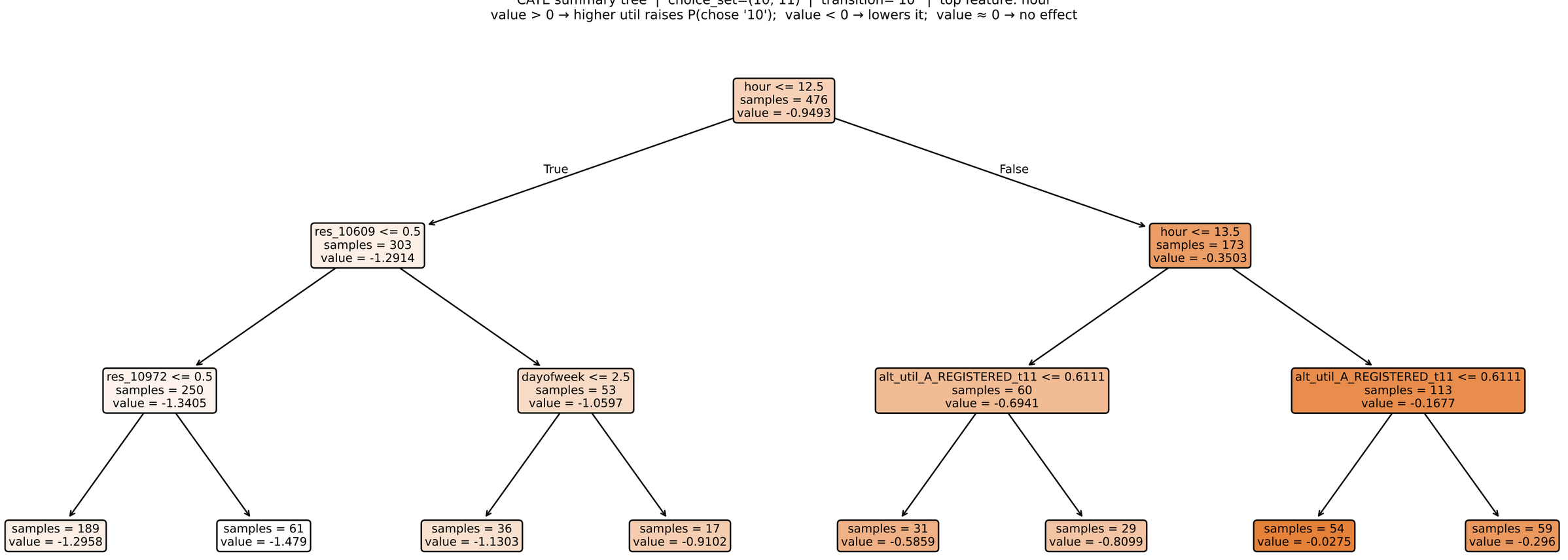

CATE summary tree | choice_set=(9, 11) | transition='11' | top feature: hour
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 13.5
samples = 1055
value = -0.5242
True
False
hour <= 9.5
samples = 843
value = -0.6083
alt_util_A_ACTIVATED_t9 <= 0.5
samples = 212
value = -0.1898
res_10809 <= 0.5
samples = 350
value = -0.6998
res_10809 <= 0.5
samples = 493
value = -0.5434
res_10138 <= 0.5
samples = 18
value = -0.353
res_10809 <= 0.5
samples = 194
value = -0.1747
samples = 306
value = -0.6719
samples = 44
value = -0.8938
samples = 417
value = -0.5237
samples = 76
value = -0.6516
samples = 11
value = -0.327
samples = 7
value = -0.3938
samples = 172
value = -0.156
samples = 22
value = -0.3208

CATE — choice_set=(10, 11) | transition='10' | n=476 | ATE=-0.9493 std=0.5173

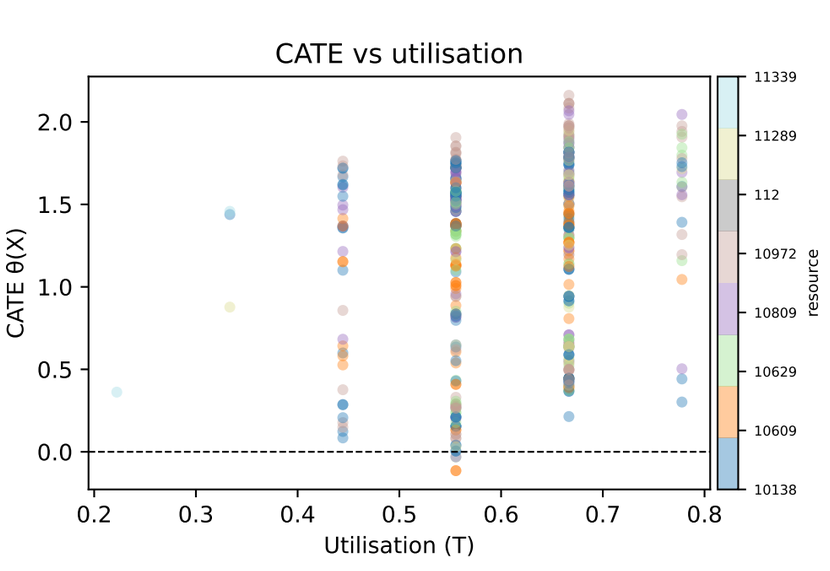


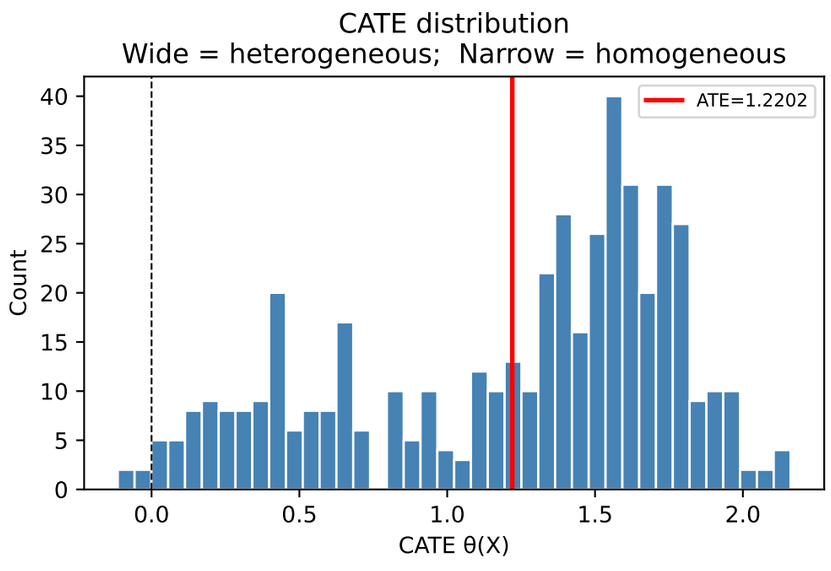


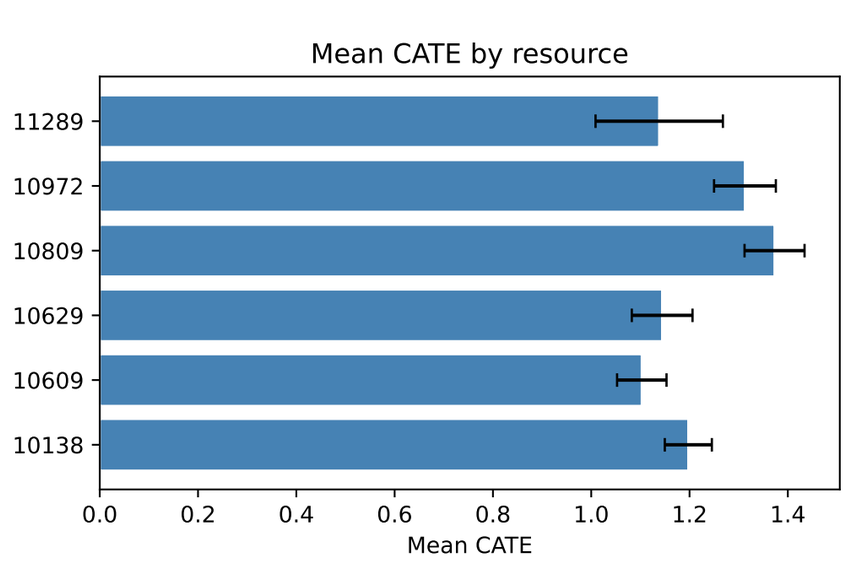


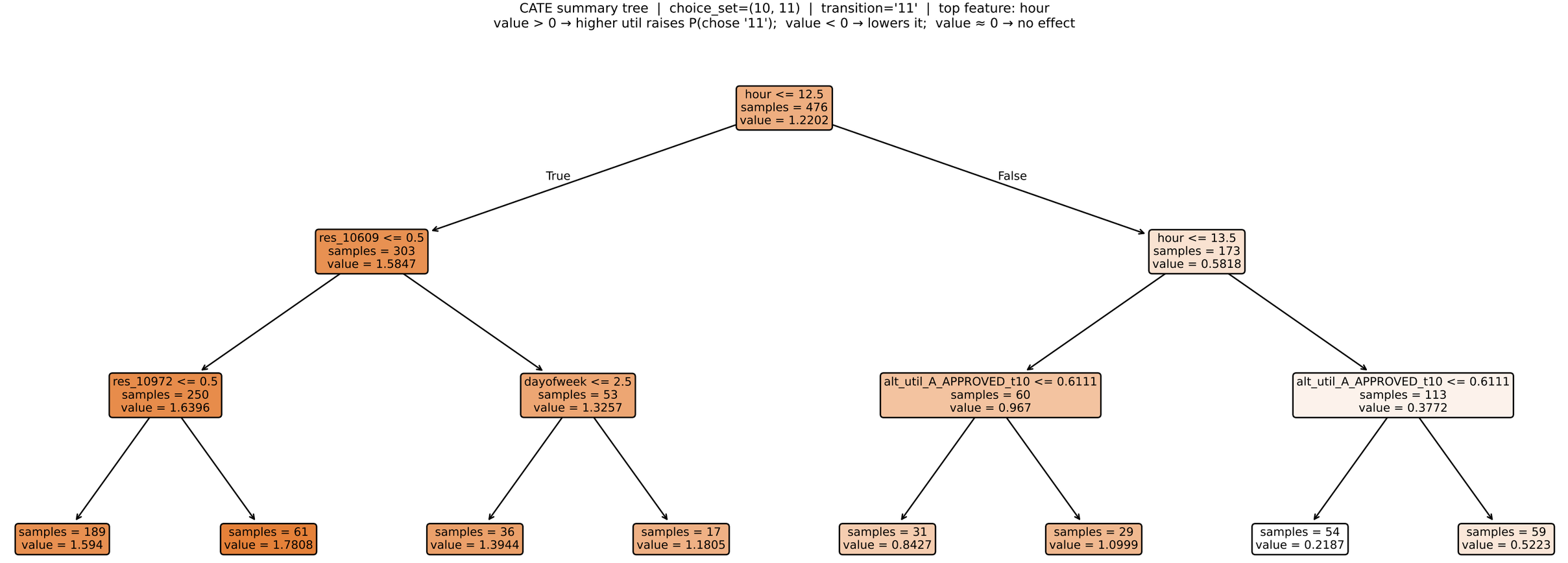

CATE summary tree | choice_set=(10, 11) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 12.5
samples = 476
value = -0.9493
True
False
res_10609 <= 0.5
samples = 303
value = -1.2914
hour <= 13.5
samples = 173
value = -0.3503
res_10972 <= 0.5
samples = 250
value = -1.3405
dayofweek <= 2.5
samples = 53
value = -1.0597
alt_util_A_REGISTERED_t11 <= 0.6111
samples = 60
value = -0.6941
alt_util_A_REGISTERED_t11 <= 0.6111
samples = 113
value = -0.1677
samples = 189
value = -1.2958
samples = 61
value = -1.479
samples = 36
value = -1.1303
samples = 17
value = -0.9102
samples = 31
value = -0.5859
samples = 29
value = -0.8099
samples = 54
value = -0.0275
samples = 59
value = -0.296

CATE — choice_set=(10, 11) | transition='11' | n=476 | ATE=1.2202 std=0.5568








CATE summary tree | choice_set=(10, 11) | transition='11' | top feature: hour
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 12.5
samples = 476
value = 1.2202
True
False
res_10609 <= 0.5
samples = 303
value = 1.5847
hour <= 13.5
samples = 173
value = 0.5818
res_10972 <= 0.5
samples = 250
value = 1.6396
dayofweek <= 2.5
samples = 53
value = 1.3257
alt_util_A_APPROVED_t10 <= 0.6111
samples = 60
value = 0.967
alt_util_A_APPROVED_t10 <= 0.6111
samples = 113
value = 0.3772
samples = 189
value = 1.594
samples = 61
value = 1.7808
samples = 36
value = 1.3944
samples = 17
value = 1.1805
samples = 31
value = 0.8427
samples = 29
value = 1.0999
samples = 54
value = 0.2187
samples = 59
value = 0.5223

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CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(8, 9, 10, 11, 12, 13)	9	12688	2.27997	412.396103	-1483.528237	-249.375494	-35.700394	202.490858	2162.018326	46.2	53.8	alt_util_A_CANCELLED_t13
(8, 9, 10, 11, 12, 13)	10	12688	-374.582741	540.435618	-3771.909569	-626.896187	-289.957738	-27.305925	1727.376903	22.3	77.7	hour
(8, 9, 10, 11, 12, 13)	11	12688	-459.401827	651.956237	-3177.08958	-850.529082	-294.528789	-4.862259	1972.601016	24.4	75.6	res_112
(8, 9, 10, 11, 12, 13)	12	12688	-0.119959	2.444875	-8.768727	-1.365874	-0.021789	1.066772	12.398527	48.9	51.1	alt_util_A_CANCELLED_t13
(8, 9, 10, 11, 12, 13)	13	12688	0.413697	2.050001	-9.979079	-0.366165	0.234159	1.367636	14.098855	62.0	38.0	alt_util_A_REGISTERED_t11
(9, 10)	9	715	-2.696851	0.489537	-3.599736	-3.039866	-2.748076	-2.387942	-1.100781	0.0	100.0	hour
(9, 10)	10	715	2.697526	0.488941	1.101782	2.384306	2.750475	3.038527	3.595585	100.0	0.0	hour
(9, 11)	9	1055	0.524168	0.212527	-0.022479	0.426704	0.544259	0.663661	1.156732	99.5	0.5	hour
(9, 11)	11	1055	-0.524243	0.212738	-1.156798	-0.663978	-0.54431	-0.426461	0.022907	0.5	99.5	hour
(10, 11)	10	476	-0.949345	0.517322	-1.795174	-1.343675	-1.094743	-0.454462	0.273408	5.0	95.0	hour
(10, 11)	11	476	1.220177	0.55685	-0.114252	0.70271	1.382278	1.637644	2.161287	99.2	0.8	hour

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md